Dashboard Server, port 29999 - 15690

Introduce all the commands that dashboard server can support. Take an example , connecting Dashboard Server with SocketTest software.

Examples are valid for:
CB2 Software version: 1.4 and upwards
CB3 Software version: 3.0 and upwards
Note that older or newer software versions may behave differently.

A Universal Robot can be controlled from remote by sending simple commands to the GUI over a TCP/IP socket. This interface is called the "DashBoard server". The server is running on port 29999 on the robots IP address. Each command should be terminated by a '\n' also called a newline.

Command	Return value	Description	Supported from	CB1	CB2	CB3	CE
load <program.urp>	"Loading program: <program.urp>" OR "File not found: <program.urp>"	Start loading the specified program. Return immediately before loading completes.	v1.4 to v3.2	x	x	x	x
		Load the specified program. Return when loading has completed.	v3.3			x	x
		Starts program, if any program is loaded and					

		robot is ready. Return immediately before starting has completed.	v1.4 to v3.2	x	x	x	x
play	"Starting program"						
		Starts program, if any program is loaded and robot is ready. Return when the program execution has been started.	v3.3			x	x
		Stops running program and return immediately.	v1.4 to v3.2	x	x	x	x
stop	"Stopped"						
		Stops running program and returns when stopping is completed.	v3.3			x	x
		Pauses the running program and return immediately.	v1.4 to v3.2	x	x	x	x
pause	"Pausing program"						

		Pauses the running program and returns when pausing is completed.	v3.3			x	x
quit	"Disconnected"	Closes connection	v1.4	x	x	x	x
shutdown	"Shutting down"	Shuts down and turns off robot and controller	v1.4	x	x	x	x
running	"Robot running: True" OR "Robot running: False" CB2: integer is returned • NO_CONTROLLER_MODE = -1 • ROBOT_RUNNING_MODE = 0 (This is "normal" mode) • ROBOT_FREEDRIVE_MODE = 1 • ROBOT_READY_MODE = 2 • ROBOT_INITIALIZING_MODE = 3 • ROBOT_SECURITY_STOPPED_MODE = 4 • ROBOT_EMERGENCY_STOPPED_MODE = 5 • ROBOT_FAULT_MODE = 6 • ROBOT_NO_POWER_MODE = 7 • ROBOT_NOT_CONNECTED_MODE = 8 • ROBOT_SHUTDOWN_MODE = 9 CB3: text is returned "Robotmode: <mode>", where <mode> is • NO_CONTROLLER • DISCONNECTED • CONFIRM_SAFETY • BOOTING • POWER_OFF • POWER_ON	Execution state enquiry	v1.6		x	x	x
robotmode		Robot mode enquiry	v1.6		x	x	x

	• IDLE • BACKDRIVE • RUNNING					
get loaded program	"Program loaded: <path to loaded program file>" OR "No program loaded"	Which program is loaded	v1.6	x	x	x
popup <popup-text>	"showing popup"	The popup-text will be translated to the selected language, if the text exists in the language file	v1.6	x	x	x
close popup	"closing popup"	Closes the popup	v1.6	x	x	x
addToLog <log-message>	"Added log message" Or "No log message to add"	Adds log-message to the Log history	v1.8.11657	x	x	x
setUserRole <role>	"Setting user role: <role>" OR "Failed setting user role: <role>"	Simple control of user privileges: controls the available options on the Welcome screen	v1.8.11657	x	x	x
isProgramSaved	"True" OR "False"	Returns the save state of the active program	v1.8.11997	x	x	x
		Returns the state of the active				

programState	"STOPPED" if no program is running OR "PLAYING" if program is running	program, or STOPPED if no program is loaded	v1.8.11997	x	x	x
polyscopeVersion	version number, like "3.0.15547"	Returns the version of the Polyscope software	v1.8.14035	x	x	x
setUserRole <role>, where <role> is • programmer • operator • none • locked	"Setting user role: <role>" OR "Failed setting user role: <role>" • programmer = "SETUP Robot" button is disabled, "Expert Mode" is available (if correct password is supplied) • operator = Only "RUN Program" and "SHUTDOWN Robot" buttons are enabled, "Expert Mode" cannot be activated • none (or send setUserRole) = All buttons enabled, "Expert Mode" is available (if correct password is supplied) • locked = All buttons disabled and "Expert Mode" cannot be activated	Control of user privileges: controls the available options on the Welcome screen. Note: If the Welcome screen is not active when the command is sent, the user privileges defined by the new user role will not be effective until the user switches to the Welcome screen.	v1.8.14035	x	x	x
power on	"Powering on"	Powers on the robot arm	v3.0		x	x
power off	"Powering off"	Powers off the robot arm	v3.0		x	x

brake release	"Brake releasing"	Releases the brakes	v3.0	x	x
safetymode	"safety mode: <mode>", where <mode> is • NORMAL • REDUCED • PROTECTIVE_STOP • RECOVERY • SAFEGUARD_STOP • SYSTEM_EMERGENCY_STOP • ROBOT_EMERGENCY_STOP • VIOLATION • FAULT	Safety mode enquiry	v3.0	x	x
unlock protective stop	"Protective stop releasing"	Closes the current popup and unlocks protective stop	v3.1	x	x
close safety popup	"closing safety popup"	Closes a safety popup	v3.1	x	x
setUserRole <role> extended with new role: • restricted	"Restricted" works like "operator" but does not give access to the move tab.		v3.1.17136	x	x
load installation <default.installation>	"Loading installation: <default.installation>" OR "File not found: <default.installation>"	Loads the specified installation file	v3.2.18654	x	x

Testing the Dashboard server:

For testing purposes a program for testing sockets can be used, like HyperTerminal, or similar. In this example the freeware program SocketTest is used.

Setup a static IP-address and subnet mask on PC, so it matches the robot, e.g.:

PC: IP-addr: 192.168.3.10 Robot: IP-addr: 192.168.3.3
Subnet: 255.255.255.0 Subnet: 255.255.255.0

In the socket test program, use the Client tab to send commands to robot:

